

Supplementary Material

S1. Evidence hierarchy

Primary evidence is the five-fold expanding temporal development evaluation. The 1,013-row Locked Registry Evaluation Set is secondary historical benchmark evidence and is not described as a newly untouched confirmatory holdout.

S2. Frozen model roles and thresholds

- CatBoost: deployment-aligned primary classifier; Platt calibrator; frozen threshold 0.1000.
- Random Forest: development PR-AUC sensitivity comparator; Platt calibrator; frozen threshold 0.1050.
- Severity point baseline: conditional median.
- Quantile/CQR engine: LightGBM quantile models at q0.025, q0.05, q0.10, q0.50, q0.90, q0.95, q0.975.

S3. Full capacity-constrained operational triage [SIMULATED SCENARIO]

K	Inspected	Strategy	Delays captured /61	High- severity /15	Delay days captured	Delay-days ratio
1%	11	Risk only	6/61	1/15	42	5.9%
1%	11	Risk x q50	9/61	8/15	318	44.4%
1%	11	Risk x q95	8/61	8/15	317	44.3%
5%	51	Risk only	16/61	2/15	92	12.8%
5%	51	Risk x q50	21/61	10/15	414	57.8%
5%	51	Risk x q95	21/61	10/15	421	58.8%
10%	102	Risk only	26/61	10/15	288	40.2%
10%	102	Risk x q50	31/61	13/15	510	71.2%
10%	102	Risk x q95	28/61	13/15	493	68.9%
20%	203	Risk only	42/61	15/15	619	86.5%

K	Inspected	Strategy	Delays captured /61	High- severity /15	Delay days captured	Delay-days ratio
20%	203	Risk x q50	36/61	15/15	574	80.2%
20%	203	Risk x q95	35/61	14/15	550	76.8%

All values above are retrospective simulated prioritization outcomes. No intervention was delivered and no causal or realized-savings claim is made.

S4. Locked CQR exact intervals

Nominal	Covered	Empirical coverage	Exact 95% Clopper- Pearson CI	Mean width	Median width
80%	43/61	70.49%	57.43-81.48%	41.04 d	32.55 d
90%	56/61	91.80%	81.90-97.28%	46.27 d	38.38 d
95%	61/61	100.00%	94.13- 100.00%	61.74 d	53.64 d

S5. Reproducibility identifiers

- Canonical dataset SHA-256:
918b992dd3e8d4b64d2a727b2c4ea607603d0c58f19484e73f7b78528c6a8673
- Final evaluation freeze SHA-256:
631F1FA4241FBE697B46D9658B8720D2CE13CFB5A9F65E32C7532F8A1F6CD21A
- Locked benchmark manifest SHA-256:
62BE80CE9BC977767BC983EDABF61DAA23609A1801251F4E93CFA9693EBDD9AB
- Frozen integrity tests: 26/26 passed.

S6. Priority-claim boundary

The literature-priority claim is limited to the joint combination of post-delivery embargoed temporal evaluation, post-hoc probability calibration, conditional severity, CQR, and explicit capacity-constrained shipment prioritization in pharmaceutical logistics. It does not claim novelty for machine learning, conformal prediction, delay-severity modeling, or prediction-to-decision integration individually.